

HK IMO Training 2020-2021
Problem Set 11
For 16 January 2020



Problem 41. Let a, b, c be positive real numbers. Prove that

$$a + \sqrt{ab} + \sqrt[3]{abc} \leq \frac{4}{3}(a + b + c).$$

Problem 42. Let a, b, c be positive real numbers. Prove that

$$\sqrt[7]{\frac{a}{b+c} + \frac{b}{c+a}} + \sqrt[7]{\frac{b}{c+a} + \frac{c}{a+b}} + \sqrt[7]{\frac{c}{a+b} + \frac{a}{b+c}} \geq 3.$$

Problem 43. Let $a_1, a_2, \dots, a_n$ be real numbers such that $a_1 + a_2 + \dots + a_n = 0$. Define

$$b_k = a_1 + a_2 + \dots + a_k$$

for $k = 1, 2, \dots, n$. It is known that $b_i(a_{j+1} - a_{i+1}) \geq 0$ for all $1 \leq i \leq j \leq n-1$. Prove that

$$\max\{|a_1|, |a_2|, \dots, |a_n|\} \geq \max\{|b_1|, |b_2|, \dots, |b_n|\}$$

Problem 44. Find all real numbers c for which there exists a function $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(f(x) + f(y)) + cxy = f(x + y)$$

for any $x, y \in \mathbb{R}$.